

Sumit Shetye

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EDUCATION

UC Santa Cruz

Master of Science in Computational Media

GPA: 4.0

Santa Cruz, CA

September 2024 – June 2025

UC San Diego

Bachelor of Science in Mathematics - Computer Science

GPA: 3.2

La Jolla, CA

September 2020 – June 2024

EXPERIENCE

Graduate Researcher

UC Santa Cruz

September 2024 – June 2025

Santa Cruz, CA

- Achieved a 99% compression ratio, reducing 5.5GB of emulator save states to just 9MB by developing a custom compression algorithm in Python using block-based deduplication and delta encoding for and lossless archival of gameplay history.
- Enabled memory-efficient replays and seamless playback with minimal storage overhead by designing a compact, chunked state stream format for “Playable Quotes,” using optimized data structures for decompression speed and fidelity.
- Boosted code reliability and long-term usability by packaging the compression toolset into a reusable Python API with a command-line interface, comprehensive test suites, documentation, and support for versioned savestream decoding.
- Demonstrated 20% better compression performance compared to mainstream tools such as 7zip by creating detailed, interactive graph visualizations in Altair, enabling clear analysis of compression ratios across multiple test datasets.

Game Development Intern

Infogen Labs, Inc.

September 2023 – December 2023

Valencia, CA

- Enhanced overall game performance and user experience by coding over 200 lines in Unity (C#), implementing responsive player movement with physics-based control and real-time scoring mechanics synced with visual feedback.
- Prototyped and launched a gesture-controlled AR version of Space Invaders by integrating Google MediaPipe’s real-time hand tracking with Unity’s input system, creating a fully playable experience that responded to intuitive user gestures.
- Participated in weekly Agile sprint reviews and collaborative playtesting sessions, contributing to the iterative refinement of gameplay mechanics and UI flow based on quantitative player feedback, in-game telemetry data, and user engagement.

Software Engineering Intern

Five Dimensions Entertainment, Inc.

January 2022 – September 2024

Los Angeles, CA

- Supported real-time event ticketing platforms by building dynamic content modules using PHP and JavaScript to push schedule changes and availability updates to thousands of concurrent users with minimal server-side latency.
- Reduced customer complaints by 30% and improved website uptime by proactively identifying, reproducing, and fixing cross-browser bugs and critical backend API issues affecting ticket flow and checkout reliability.

PROJECTS

Escrow Timeline Calculator | *Typescript, React, Node.js*

June 2025 – Present

- Simplified complex real estate transactions for users by building a full-stack web tool that calculates custom escrow timelines, using TypeScript and React to guide users through configurable input flows with real-time feedback.
- Developed RESTful backend services in Node.js to generate data-driven milestone schedules for escrow transactions, with a roadmap to integrate Google Calendar syncing, PDF timeline export, and automated push or email notifications.
- Built the app using Docker containers for consistent development environments, with a CI pipeline in place as well as AWS and Kubernetes deployment architecture under development to simulate enterprise cloud workflows.

Artist Recommender | *Python, HTML, Javascript, Gemini API*

January 2025

- Designed and deployed a personalized artist recommendation system that achieved 10% higher match accuracy than Spotify and Apple Music baselines, using prompt engineering with Google’s Gemini LLM APIs to extract patterns from user input.

FutureNOW! | *Javascript, HTML, CSS, Node.js*

March 2023 – June 2023

- Boosted team productivity by 35% through the implementation of automated CI/CD pipelines using GitHub Actions, integrating linting, code formatting, and test verification workflows to ensure faster review cycles.
- Improved application robustness and long-term maintainability by writing over 20 unit, integration, and end-to-end tests using JUnit (backend), Jest (frontend), and Cypress (browser automation), achieving over 90% coverage.

TECHNICAL SKILLS

Languages: Python, C++, Javascript, Typescript, C, C#, HTML/CSS, Matlab, SQL, MS SQL, MySQL, Bash

Frameworks: React, Node, JUnit, WordPress, FastAPI

Developer Tools: Linux, AWS, Git, Kubernetes, CI/CD Pipeline, Docker, Jupyter Notebook, VS Code, Visual Studio

Libraries: pandas, NumPy, Matplotlib, Google Generative AI, MediaPipe, Altair